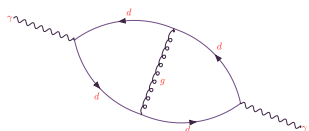


Tested diagrams and generation times

L localized 3D E explicit 3D P projected 4D

Each card shows passed UV rays per route and generation times in seconds, ordered L / E / P.

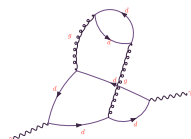


1 Double triangle

2-loop U control

UV 3/3 · 24/24 rays

7.32 / 5.97 / 2.22 s

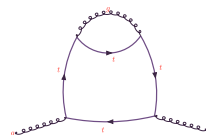


2 DT + bubble

Route + soft checks

UV 3/3 · 196/196 rays

851.6 / 707.4 / 729.0 s†

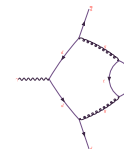


3 Nested gluon SE

Appendix B.1

UV 3/3 · 21/21 rays

9.88 / 8.98 / 5.42 s

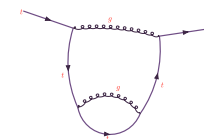


4 Top-bubble vertex

H gains two soft powers

UV 3/3 · 27/27 rays

9.48 / 8.78 / 1.97 s

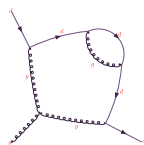


5 Nested top SE

Nested soft parts

UV 3/3 · 21/21 rays

2.62 / 2.57 / 0.92 s



6 DGSE

Selected soft checks pass

UV 3/3 · 27/27 rays

3.62 / 3.22 / 0.92 s

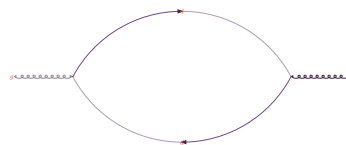


Quark self-energy

1 loop

UV 3/3 · 2/2 rays

0.62 / 0.62 / 0.62 s

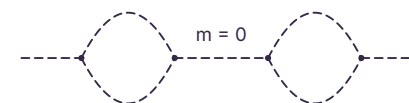


Gluon self-energy

1 loop

UV 3/3 · 2/2 rays

0.62 / 0.62 / 0.62 s

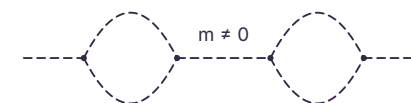


Disconnected soft parts

Massless bridge

UV 3/3 · 12/12 rays

3.72 / 3.57 / 0.82 s



Consecutive soft parts

Massive bridge

UV 3/3 · 12/12 rays

3.77 / 3.57 / 0.77 s

$$H_d[F] = S_{d-1}[F] + U_d[F - S_{d-1}[F]]$$

This defines H for $d > 0$; $d = 0$ uses U. Each parent acts on its completed children.

The child-only top-bubble control passes 13/27 UV rays; its other sectors are unsubtracted. Generation takes 3.47 / 3.22 / 1.27 s.

Soft and pointwise checks cover selected cases, as shown on slide 3.

These are single historical runs, including startup. Generation was measured on 25 Sep; †27 Sep runs retained numerator functions.

Expression changes speed up algebra and evaluation

We keep functions and powers intact, factor the remaining expressions, and share repeated scalar terms.

Change	Measured improvement
Compact energies and Taylor reuse	Time for the same seven Taylor operations fell from 398 s to 19.6 s — about 20× faster . Peak memory fell from 9.18 GiB to 0.11 GiB .
Symbolica Horner fix	In the 1,000-group scalar test, Horner time fell from 619 ms to 6.26 ms — about 99× faster . Peak memory fell from 797.5 MiB to 19.0 MiB .
Numerator inlining DT + bubble, full H	Evaluation time fell from 24.38 to 2.45 ms in Double (about 10× faster) and from 2.65 to 0.170 s at 1000 bits (about 16× faster). Explicit 3D sums all 102 orientations at the same point. Generation took 12m49s and peaked at 3.03 GiB .

The GL262 evaluator build peaked at 30.26 GiB

One build creates **135.7 million operations**, then keeps **3.43 million** after optimization. These counts describe temporary growth, not a measured speedup.

Turning off common-pair elimination did not lower the memory peak.

Direct sharing helps in the reproducer

In the standalone Rust test, sharing built the same 1,919-operation evaluator in **1.49 ms** instead of **4.24 s**. Peak memory fell from **67.7 MiB** to **5.92 MiB**.

Direct sharing has not been implemented for GL262.

On main, explicit U generation took 25m51s and peaked at 80.9 GiB. The other two routes hit the 100 GiB cap; the time limit was 8 h. These runs used historical builds from 23 Sep–1 Oct. Speedup claims use paired measurements; full run details are in the source.

All ten diagrams pass the UV checks

All three routes pass UV checks for all ten fully subtracted diagrams on slide 1.
These runs were made on 25–28 Sep. The child-only control is excluded.

Soft and pointwise checks cover selected cases.

Diagram 2 · three routes agree

198 comparisons agree
We tested 33 points, two rotations and three route pairs at 1000-bit precision.
The largest relative complex difference was 3.514×10^{-257} .

UV: All 196 rays pass in each route at 25 scales.

H improves central soft scaling by two powers

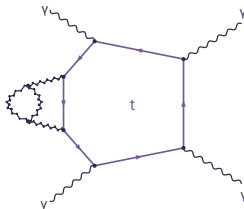
$$|I_H| \sim \lambda^{-1}, \quad |I_U| \sim \lambda^{-3}$$
$$\lambda^3 |I_H| \sim \lambda^2 \quad (\text{U: plateau})$$

With inlining, these powers were measured on three central-gluon rays.
Each value sums 102 orientations.

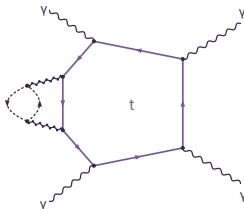
We have not tested individual orientations or proved algebraic equality.

These runs were made on 25–30 Sep. Values sum orientations; integrated and threshold counterterms were disabled.

Four photons · only projected 4D was tested



GL262 · gluon bubble
Graph factor -2



GL256 · ghost bubble
Graph factor -4

Graph	Generate	Peak RSS	Weighted soft power
GL262	672.20 s	30.56 GiB	2.000005
GL256	306.67 s	15.60 GiB	1.999987

The original [4,9,12] basis works with compatible sub-LMBs and inlining. Each graph passes four selected UV limits.
At three points per graph, Double and Arb agree to 10^{-13} . We have not run a full UV scan or compared routes.

CI found three code defects that still need fixes

2,942

passed

34

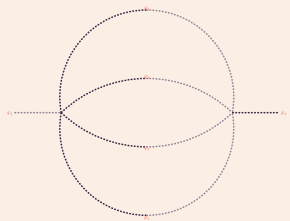
failed · all investigated

308

skipped

Clippy, formatting, CI graph checks and 47 doctests pass.

1 failure · momentum routing



3-loop banana

A missing boundary endpoint gives the wrong loop momentum.

At $t = 1$:

$$Q_2^{\text{routed}} - Q_2^{\text{physical}} = Q_4 \neq 0$$

The error occurs before Taylor/CFF. The routes differ by 24%; neither has been validated.

3 failures · FORM output

Vakint writes $\frac{i}{16}$ as $(*i_/16)$, which FORM rejects. This blocks the later checks.

1 failure · rational cancellation

Taylor completes, but rational cancellation then crashes on a valid floating $\zeta(3)$ coefficient.

We must check the coefficient type before cancellation and keep the ε^2 terms.

29 other test failures

- 15 The snapshots changed formatting; the expressions are equal.
- 7 The documentation checks fail.
- 7 The failures involve parsing, scope and normalization checks.

The audit made no code or snapshot changes. Later assertions in two tensor tests remain unchecked.

Next: Fix the three defects and rerun the unchanged tests before resuming exact overlap checks.

We still need to resolve the U mismatch with main (ratio 2.25539), prove overlap-pole cancellation, and validate local OS and integrated amplitudes.

NixCI 652910c3 · audit 802dd642 · Symbolica 578dfcb.